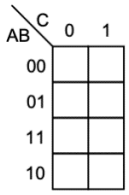
Tutorial 07 - Karnaugh Maps

1. A Karnaugh map is nothing more than a special form of truth table, useful for reducing logic functions into minimal Boolean expressions.
   Here is a truth table for a specific three-input logic circuit:

A	B	C	Out
0	0	0	1
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	0
1	1	1	0

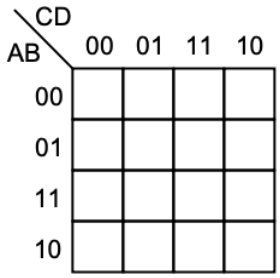
   Complete the following Karnaugh map, according to the values found in the above truth table:



2. Here is a truth table for a specific four-input logic circuit:

A	B	C	D	Out
0	0	0	0	0
0	0	0	1	1
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	1
0	1	1	0	0
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	1
1	1	1	0	0
1	1	1	1	1

Complete the following Karnaugh map, according to the values found in the above truth table:



3. Use a Karnaugh map to generate a simple Boolean expression
    a. $Z = \overline{A}\,\overline{B}\,\overline{C} + \overline{A}B + AB\overline{C} + AC$
    b. $Z = \overline{A}B + B\overline{C} + BC + A\overline{B}\,\overline{C}$

4. A manufacturing plant needs to have a horn sound to signal quitting time. The horn should be activated when either of the following conditions is met:
    a. It's after 5 o'clock and all machines are shut down
    b. It's Friday, the production run for the day is complete, and all machines are shut down.

    Design a logic circuit that will control the horn. (Hint: Use four logic input variables to represent the various conditions; for example, input A will be HIGH only when the time of day is 5 o'clock or later.) Follow these steps to design the circuit:
    a. Identify the input and outputs
    b. Construct the truth table
    c. Obtain the logic expression (SOP)
    d. Minimize it using a K-map and obtain the minimized logic expression from the K-map
    e. Implement the solution using logic gates

5. Simplify using K-maps:
    a. $f(A,B,C,D)=\Sigma\,(1,3,5,6,7,11,13,14)$
    b. $f(A,B,C) = \Sigma\,(1,3,5,6,7)$
    c. $f(A,B,C,D) = \Sigma\,(2,3,4,5,7,8,10,13,15)$
    d. $f(A,B,C,D)= \Sigma(0,5,7,8,10,12,14,15)$
    e. $f(A,B,C,D) = \Sigma\,(1,2,4,6,9)$

Extra Questions:

1. Use Karnaugh maps to find the minimal form for each expression.
    a. xy + xy’
    b. xy + x’y + x’y’
    c. xy’ + x’y’
    d. xyz + xyz’ + xy’z
    e. xyz’ + xy’z + xy’z’ + x’yz + x’yz’ + x’y’z
    f. xyz + xyz’ + x’yz + x’y’z
    g. xyz + xyz’ + xy’z + xy’z’ + x’y’z

2. Use a Karnaugh map to generate a simple Boolean expression for this truth table:

A	B	C	Output
0	0	0	0
0	0	1	0
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

3. Use a Karnaugh map to generate a simple Boolean expression for this truth table, and draw a gate circuit equivalent to that expression:

A	B	C	D	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

4. Use a Karnaugh map to generate a simple Boolean expression for this truth table, and draw a gate circuit equivalent to that expression.

A	B	C	D	Output
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	1
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	1
1	1	1	1	1

5. Use a Karnaugh map to generate a simple Boolean expression for this truth table and draw a gate circuit equivalent to that expression.

A	B	C	D	Output
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	1
0	1	1	1	0
1	0	0	0	1
1	0	0	1	0
1	0	1	0	0
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	1
1	1	1	1	0